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1. 5 Vs of Big Data with Flipkart Case Study

Introduction

Big Data refers to extremely large and complex datasets that cannot be efficiently processed using traditional data processing tools. Organizations such as Flipkart, Amazon, Netflix, and Google generate enormous amounts of data every second. To understand the characteristics of Big Data, the concept of the **5 Vs** is commonly used.

1. Volume

Volume refers to the massive amount of data generated and stored by an organization.

Flipkart Example

Flipkart handles millions of customer transactions every day. Data is generated from:

- Product searches
- Customer orders
- Payment transactions
- Product reviews
- Wishlist activities

During events such as the Big Billion Days Sale, the amount of generated data increases significantly.

Importance

Large data volumes help businesses identify trends, customer preferences, and purchasing behavior.

2. Velocity

Velocity refers to the speed at which data is generated, transmitted, and processed.

Flipkart Example

When customers browse products, place orders, track shipments, or make payments, data is generated continuously in real time.

Examples:

- Real-time order updates
- Live inventory tracking
- Instant payment confirmations
- Recommendation engines

Importance

Fast processing enables businesses to make immediate decisions and provide better customer experiences.

3. Variety

Variety refers to different types of data formats.

Flipkart Example

Flipkart manages:

- Structured Data (Orders, Customer Details)
- Semi-Structured Data (JSON logs)
- Unstructured Data (Images, Videos, Reviews)

Importance

Combining different data types provides deeper business insights.

4. Veracity

Veracity refers to the quality, reliability, and accuracy of data.

Flipkart Example

Challenges include:

- Fake reviews
- Duplicate records
- Incorrect customer information
- Invalid transactions

Flipkart uses validation techniques and machine learning algorithms to improve data quality.

Importance

Accurate data leads to accurate business decisions.

5. Value

Value refers to extracting meaningful business insights from data.

Flipkart Example

Flipkart uses data to:

- Recommend products
- Forecast demand
- Optimize inventory
- Improve marketing campaigns

Importance

Data becomes useful only when it generates business value.

Conclusion

The 5 Vs help organizations understand and manage Big Data effectively. Companies like Flipkart use Big Data technologies to improve customer satisfaction, operational efficiency, and business growth.

2. Transformations vs Actions in PySpark

Introduction

PySpark operations are categorized into:

1. Transformations
2. Actions

Understanding the difference is essential because Spark follows a **Lazy Evaluation** model.

What are Transformations?

Transformations create a new DataFrame or RDD from an existing one.

They do not execute immediately.

Common Transformations

```
df.select("name")
```

```
df.filter(df.salary > 50000)
```

```
df.groupBy("department")
```

```
df.withColumn("bonus", df.salary * 0.1)
```

Example

```
employees = df.filter(df.salary > 50000)
```

At this stage Spark only records the operation.

No actual computation occurs.

What are Actions?

Actions trigger execution and produce results.

Common Actions

```
show()
```

```
count()
```

```
collect()
```

```
first()
```

```
take()
```

Example

```
employees.show()
```

Spark executes all pending transformations and returns the result.

Comparison Table

Feature	Transformations	Actions
Execution	Lazy	Immediate
Result	New DataFrame/RDD	Actual Output
Processing	Not Executed	Executed
Example	<code>filter()</code>	<code>show()</code>

Why Lazy Evaluation?

Benefits:

- Query Optimization
- Reduced Computation
- Better Resource Utilization
- Faster Execution

Conclusion

Transformations define operations while Actions execute them. Together they form the foundation of PySpark processing.

3. Top 5 Cities by Revenue Using PySpark DataFrame API

Objective

The objective is to identify the top-performing cities based on total revenue.

Sample Dataset

City	Revenue
Chennai	5000
Coimbatore	7000
Chennai	3000

PySpark Query

```
from pyspark.sql.functions import sum
```

```
top_cities = (  
    df.groupBy("city")  
      .agg(sum("revenue").alias("total_revenue"))  
      .orderBy("total_revenue", ascending=False)
```

)

```
top_cities.show(5)
```

Explanation

groupBy()

Groups records city-wise.

```
groupBy("city")
```

agg()

Calculates aggregated revenue.

```
agg(sum("revenue"))
```

orderBy()

Sorts cities by highest revenue.

```
orderBy("total_revenue", ascending=False)
```

show(5)

Displays only the top 5 cities.

```
show(5)
```

Business Use Cases

- Identify profitable regions
- Improve marketing strategies
- Optimize inventory allocation
- Support decision-making

4. Parquet vs CSV: Performance and Storage Comparison

Introduction

Data storage format significantly affects performance and storage efficiency.

Two commonly used formats are:

- CSV
- Parquet

CSV

CSV stands for Comma Separated Values.

Example:

ID,Name,Revenue

1,Alex,5000

2,John,7000

Advantages

- Easy to read
- Human readable
- Supported everywhere

Disadvantages

- Large storage size
- Slow query performance
- No compression

Parquet

Parquet is a columnar storage format optimized for analytics.

Advantages

- Compression support
- Faster queries
- Better storage efficiency
- Schema preservation

Disadvantages

- Not human readable
- Requires specialized tools

Comparison Table

Feature	CSV	Parquet
Storage	High	Low
Compression	No	Yes
Query Speed	Slow	Fast
Schema Support	No	Yes
Big Data Usage	Limited	Excellent

PySpark Commands

Save as CSV

```
df.write.csv("sales_csv")
```

Save as Parquet

```
df.write.parquet("sales_parquet")
```

Conclusion

Parquet is preferred in Big Data systems because it provides better performance and storage optimization.

5. Medallion Architecture (Bronze, Silver, Gold)

Introduction

Medallion Architecture is a multi-layer data architecture used in Data Engineering to improve data quality and support analytics.

It consists of:

1. Bronze Layer
2. Silver Layer
3. Gold Layer

Bronze Layer

Purpose

Stores raw data exactly as received.

Characteristics

- Original data
- No cleaning
- Historical storage

Example

```
bronze_df = spark.read.csv(  
    "sales.csv",  
    header=True,  
    inferSchema=True  
)
```

Silver Layer

Purpose

Stores cleaned and transformed data.

Activities

- Remove duplicates
- Handle missing values
- Standardize formats

Example

```
silver_df = bronze_df.dropDuplicates()  
silver_df = silver_df.na.fill(0)
```

Gold Layer

Purpose

Stores business-ready aggregated data.

Example

```
from pyspark.sql.functions import sum
```

```
gold_df = silver_df.groupBy("city") \
    .agg(sum("revenue")
    .alias("total_revenue"))
```

Architecture Flow

Raw Data

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Bronze Layer

|



Silver Layer

|



Gold Layer

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Dashboards & Reports

Advantages

- Better Data Quality
- Easy Maintenance
- Scalability
- Faster Analytics

- Improved Governance